

Neha Valuthundil Shaji

BSC (HONS) CREATIVE COMPUTING STUDENT | MACHINE LEARNING & DATA SCIENCE ENTHUSIAST

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PROFESSIONAL OBJECTIVE

Undergraduate student with a strong interest in applied AI research and hands-on experience in machine learning, computer vision, and data-driven experimentation. Skilled in analyzing model performance, designing experiments, and deriving actionable insights. Seeking participation in MBZUAI UGRIP to contribute to faculty-led AI research projects, collaborate with talented peers, and further develop practical research skills in machine learning and data-driven systems.

EDUCATION

BSc (Hons) Creative Computing, Bath Spa University, Ras Al Khaimah

(2023 - 2026)

CGPA: 3.3 / 4.0 (Years 1-2 completed)

Machine Learning (Year 3): 70% (First-Class)

Relevant Coursework: Machine Learning, Research Project

Award: Academic Excellence Award (Year 2, Semester 2)

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript.
- **Machine Learning & Data Tools:** NumPy, Pandas, Scikit-learn, TensorFlow, Keras, Matplotlib, Seaborn, Jupyter Notebook, Git, Gradio
- **Machine Learning Methods :** Regression, Classification, CNN, Transfer Learning, Data Preprocessing, Cross-Validation, Model Evaluation, Statistical Analysis, Experimental Design

ACADEMIC & RESEARCH PROJECTS

• Weather Image Classification (CNN & Transfer Learning)

Investigated transfer learning for small-scale weather image classification using MobileNetV2 with augmentation and performance evaluation; deployed interactive inference interface demonstrating model generalisation.

Kaggle : <https://www.kaggle.com/code/neha005/weather-image-classification-ml-project>

• Online Purchase Prediction (Classification)

Performed EDA, imbalance mitigation, and comparative benchmarking across multiple classifiers to analyse trade-offs between model complexity and predictive accuracy.

Kaggle : <https://www.kaggle.com/code/neha005/02-classification-task>

• Sports Car Price Prediction (Regression)

Conducted data cleaning and feature preparation (~1k records) and evaluated regression models using cross-validation to assess predictive stability and feature influence.

Kaggle : <https://www.kaggle.com/code/neha005/01-regression-task>

• Immersive 360° Educational Storytelling (Research Project)

Designed and analysed survey-based study (n=100) exploring relationships between immersive interaction, engagement, and perceived learning outcomes.

Link:- https://docs.google.com/document/d/1BE_JSxqosQD70vdapbvh0fLbgYH2i325/edit?pli=1

RELEVANT CERTIFICATIONS

- **IBM:** Python for Data Science
- **Kaggle:** Intro & Intermediate Machine Learning; SQL for Data Science
- **LinkedIn Learning:** Probability Foundations for Data Science

RESEARCH INTERESTS

- Machine Learning, Computer Vision, Data Science, and Human-AI Interaction with emphasis on applied AI experimentation and real-world data driven research.